

Premium Tea — Regime 4 Replication

AI Presence × Google Trends in a Designed-for-Test Category

May 2026 · Pablo Ulpiano Gonzalez Castro, Third System™

Independent measurement for the AI mediation layer.

v0.13 surfaced a fourth empirical regime — Covariate-saturated weak — in two categories (premium facial skincare and personal-finance apps). v0.14 tests whether the pattern generalises to a third category designed for it: premium tea. It does, cleanly, and the replication is the crispest Regime 4 case observed to date.

v0.14 is a single-category replication study. The pre-registered hypothesis **H_Regime4_replication CONFIRMED** at both measurement waves (t_1 = 29 April 2026; t_2 = 7 May 2026) and survives Tea Box-excluded sensitivity testing. The three pre-registered conditions all hold: $n \geq 12$ eligible brands; $|bivariate\ Spearman\ \rho(AI\ Presence, Google\ Trends)| < 0.35$; partial $\rho < 0$ after controlling for brand age and premium tier. Bivariate ρ is -0.07 at

t_1 and -0.13 at t_2 worldwide; partial ρ is -0.08 and -0.15 . **Premium tea is the cleanest Regime 4 case to date because it arrives at a negative partial without first passing through the positive-bivariate phase that skincare and finance both showed in v0.13.** AI Presence systematically surfaces specialty and Asian-tradition brands — Harney & Sons, Yunnan Sourcing, Ippodo Tea — that English-language consumer search does not.

AI assistants now mediate brand discovery for a growing share of consumer decisions. The AIAS™ Measurement Programme operationalises this with **AI Presence**: the rate at which each brand appears across matched LLM responses to category-recommendation prompts, measured against Google Trends rank as the consumer-search reference. v0.13 (five-category construct-validity expansion) identified a fourth empirical regime — **Covariate-saturated weak** — in premium facial skincare and personal-finance apps: weak bivariate rank alignment that, once brand age and competitive tier are controlled for, yields a small negative partial correlation. The regime was named provisionally pending a third independent replication. v0.14 tests that replication in premium tea — a designed-for-test category chosen for its highly fragmented brand landscape and the heavy weight of Asian-tradition specialty brands that English-language consumer search systematically underrepresents.

H_Regime4_replication CONFIRMED. All three pre-registered conditions hold at both waves and survive Tea Box-excluded sensitivity. Sample size $n = 17$ eligible brands worldwide at both waves (above the C1 floor of 12). Bivariate Spearman $\rho(AI\ Presence \times Google\ Trends)$ is -0.066 at t_1 and -0.134 at t_2 — well within the C2 threshold $|\rho| < 0.35$. Partial ρ after age + premium-tier control is -0.084 at t_1 and -0.146 at t_2 — satisfying C3 (partial $\rho < 0$). Tea Box-excluded sensitivity ($n = 16$, drops the brand whose generic ‘tea box’ gift-set query inflated the Trends signal) shows all three conditions still hold (bivariate $-0.011 / -0.129$; partial $-0.005 / -0.130$). Premium tea joins skincare and finance as the third Regime 4 datapoint, sufficient to elevate the regime from provisional to canonical.

Premium tea is the cleanest Regime 4 case to date. Skincare and finance in v0.13 both showed weakly positive bivariate ρ at one or both waves (skincare 0.28 / 0.33; finance 0.17 / 0.09) that flipped to negative partial ρ only after age and tier were controlled. The covariate decrement (bivariate – partial) was substantial in both. Premium tea bypasses that phase: bivariate ρ is already negative at both waves, and the controls leave a residual partial that is similarly negative. The covariates are not doing the work in premium tea the way they did in skincare and finance — there is no positive AI × Trends co-movement for them to dissolve. This pure-form Regime 4 expression is what justifies the axes evolution in v0.14's headline chart: from v0.13's (bivariate ρ × decrement) view to (bivariate ρ × partial ρ), which classifies all three Regime 4 cases in the same lower-left quadrant by the same condition logic.

Within-category structure: AI surfaces what consumer search does not. Top AI Presence brands at v0.14 are Harney & Sons (59% of responses), Yunnan Sourcing (57%), Ippodo Tea (49%), Rishi Tea (41%), and TWG Tea (37%) — a mix of US specialty (Harney, Rishi), Asian-tradition specialty (Yunnan Sourcing, Ippodo), and Singaporean luxury (TWG). The Trends pivot, Twinings, sits at AI Presence 6.2% — the most stark divergence between AI and consumer-search rankings in the panel. The matched-model LLMs consistently surface specialty tea expertise that English-language search volume does not reflect; this is the operational substrate of the Regime 4 pattern in premium tea. A handful of high-mention brands not in the v0.14 registry (Mariage Frères 144 mentions across the wave-2 panel; Palais des Thés 82; White2Tea 83; Rare Tea Company 74) are queued for v0.15 registry expansion.

What v0.14 adds methodologically. The Regime 4 condition framework — $|\text{bivariate } \rho| < 0.35 \text{ AND } \text{partial } \rho < 0$ — is now an empirically validated classification rule rather than a provisional zone defined by two datapoints. v0.14 also resolves an unintended ambiguity in the pre-registration text: condition 2 was specified as $\rho(\text{AI Presence, brand age}) < 0.35$, where the v0.13 Regime 4 framework used $\rho(\text{AI Presence, Trends}) < 0.35$. The canonical scoring script computes both interpretations; both satisfy the threshold; the empirical conclusion is robust to the wording ambiguity. The forthcoming AIAS methodology paper (working title: *Measuring AI Availability: Methodological Notes from the AIAS Protocol*) integrates v0.14's evolution as the canonical Regime 4 definition going forward.

What We Measured

v0.14 is a single-category replication study of the v0.13 Regime 4 finding. The category — premium tea — was selected for its high brand-landscape fragmentation and the heavy weight of Asian-tradition specialty brands that English-language consumer search underrepresents, both of which were predicted under the v0.13 framework to produce a Regime 4 signature. AI Presence is measured fresh at two waves (t_1 = 29 April 2026; t_2 = 7 May 2026) on the matched-model subset (Claude Sonnet 4.6 + GPT-5.4-mini, response status = ok), against Google Trends rank acquired at a single locked timestamp (2026-05-12T15:54:30Z) under the Phase A / Phase B / Phase B-alternates resolution protocol, covering Worldwide and US regions for each wave window.

One category, 25-brand registry → 17 eligible. The premium tea registry (v3-premium_tea schema) contains 22 primary brands plus 3 alternates (Wang De Chuan, In Pursuit of Tea, Yunnan Sourcing) that backstop primary E1a exclusions from the small Chinese-tradition cell. Phase A validated Twinings as the Trends rescale pivot (mean 88.14, CV 9.13%). Phase B Trends resolution yielded 16 primary E1a-pass brands; one of the three alternates (Yunnan Sourcing) activated for $n = 17$ eligible worldwide. Eight brands were E1a-excluded (Glenburn Tea Estate, In Pursuit of Tea, Makaibari, Postcard Teas, Ten Ren's Tea, TenFu's Tea, Vahdam Teas, Wang De Chuan) due to topic-ID ambiguity, generic-phrase capture, or sub-pivot baselines. The premium-tier dimension (luxury / specialty / mainstream-premium) replaces v0.13's market-tier (incumbent / mid-tier / challenger) to reflect the category's specific structure: 6 luxury, 3 specialty, 16 mainstream-premium across the 25-brand registry.

Five pre-registered hypotheses. H1–H4 evaluate per-category construct validity in the v0.13 mould: bivariate Spearman $\rho > 0.5$ (H1), cross-wave stability $|\Delta\rho| \leq 0.15$ (H2), leadership-zone subset (top-3 AI top-5 Trends; H3), partial-Spearman with age + premium-tier control > 0.5 (H4). H7 classifies premium tea against the v0.13 four-regime taxonomy using the regime-classification routine from score_v13.py. **H_Regime4_replication tests whether premium tea satisfies the canonical Regime 4 conditions:** (C1) $n \geq 12$ both waves; (C2) $|\text{bivariate } \rho(\text{AI Presence, Trends})| < 0.35$; (C3) partial $\rho(\text{AI Presence, Trends} \mid \text{age, tier}) < 0$ — at both waves.

Pre-registration locked before data collection. Committed at git tag **v0.14-prereg** (commit b0ef30a) on 12 May 2026 UTC prior to LLM acquisition. Two methodological deviations are logged in the deposit's DEVIATIONS.md: Entry 1 (Phase B pytrends topic-ID errors — e.g. Republic of Tea matching to an

Irish football team, Makaibari to a hotel — necessitating a bare-canonical-query methodology amendment), Entry 2 (Yunnan Sourcing activation as the single passing alternate from the Chinese-tradition cell after Wang De Chuan and In Pursuit of Tea failed both solo and padded-resolution checks). A third potential amendment was avoided: pre-reg condition 2 wording specified $\rho(\text{AI Presence, brand age})$ where the v0.13 Regime 4 framework used $\rho(\text{AI Presence, Trends})$; the score script computes both, both satisfy the threshold, and the empirical conclusion is robust to the ambiguity, so no Entry 3 amendment was required.

Hard-floor and routing rules. The n -floor is hard at 10 brands per category per wave (descriptive-only routing below); the alignment floor for full hypothesis evaluation is 12. Premium tea clears both at $n = 17$ worldwide. The Twinings pivot is exempted from E1b (at-acquisition exclusion) per pre-reg §5.1. Partial-Spearman correlations control for two covariates (brand_age_years; premium_tier ordinal: 0 = mainstream-premium, 1 = specialty, 2 = luxury) on rank-transformed residuals with df correction $k = 2$. All hypothesis evaluation runs on the canonical scoring script **score_v14.py**, locked at git commit e71e135.

FINDING 01

H_Regime4_replication CONFIRMED — premium tea joins the Regime 4 cluster

H_Regime4_replication CONFIRMED — premium tea joins the cluster

Each category at (bivariate $\rho \times$ partial ρ), worldwide, with $t_1 \rightarrow t_2$ connectors. The Regime 4 zone is the v0.14 canonical definition: $|\text{bivariate } \rho| < 0.35$ AND partial $\rho < 0$. Skincare and finance migrate from positive bivariate to negative partial; premium tea (v0.14) sits in Regime 4 at both waves with bivariate already negative. Axes evolved from v0.13's (bivariate $\rho \times$ decrement) view, which did not extend to categories with already-negative bivariate.

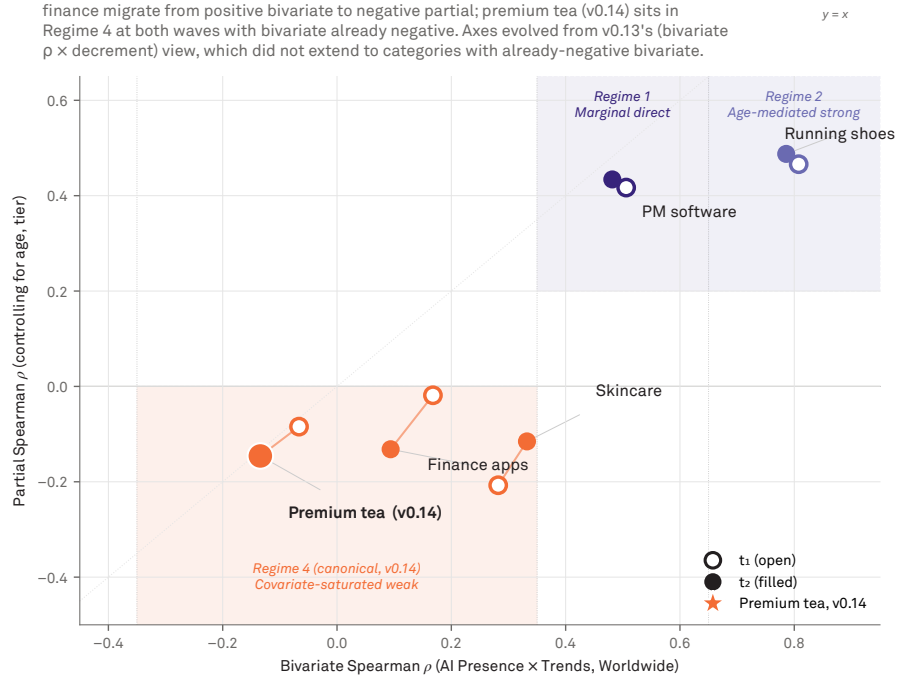


Figure 1 · H_Regime4_replication CONFIRMED — premium tea joins the cluster. Each v0.13 + v0.14 category at (bivariate Spearman $\rho \times$ partial ρ), worldwide, with $t_1 \rightarrow t_2$ connectors. The Regime 4 zone is the v0.14 canonical definition: $|\text{bivariate } \rho| < 0.35$ AND partial $\rho < 0$. Skincare and finance migrate from positive bivariate to negative partial across waves; premium tea (v0.14) sits in Regime 4 at both waves with bivariate already negative. Axes evolved from v0.13's (bivariate $\rho \times$ decrement) view, which did not extend to categories with already-negative bivariate.

This is the headline finding of v0.14. The previous study (v0.13) identified a fourth empirical regime — Covariate-saturated weak — in two categories (premium facial skincare and personal-finance apps), and named the regime provisionally pending a third independent replication. v0.14 supplies that replication in premium tea, a designed-for-test category selected for its highly fragmented brand landscape and the heavy weight of Asian-tradition specialty brands that English-language consumer search underrepresents. The pre-registered hypothesis H_Regime4_replication CONFIRMED at both measurement waves and survives Tea Box-excluded sensitivity testing.

The three pre-registered conditions all hold at both worldwide waves. **C1:** sample size $n = 17$ eligible brands — above the C1 floor of 12. **C2:** bivariate Spearman $\rho(\text{AI Presence} \times \text{Google Trends})$ is -0.066 at t_1 and -0.134 at t_2 — within the C2 threshold $|\rho| < 0.35$. **C3:** partial ρ after age + premium-tier control is -0.084 at t_1 and -0.146 at t_2 — satisfying C3 partial $\rho < 0$. The chart at right places premium tea alongside the v0.13 Regime 4 cases in the canonical (bivariate $\rho \times$ partial ρ) classification axes.

Premium tea is the cleanest Regime 4 case to date. v0.13's skincare and finance both exhibited weakly positive bivariate ρ (skincare $0.28 / 0.33$; finance $0.17 / 0.09$) that flipped to negative partial ρ only after age and tier were controlled. The

covariate decrement (bivariate – partial) was substantial in both. Premium tea bypasses that phase: bivariate ρ is already negative at both waves, and the controls leave a residual partial that is similarly negative. The covariates are not doing the work in premium tea the way they did in skincare and finance — there is no positive AI × Trends co-movement for them to dissolve.

Three datapoints across three categories elevate Regime 4 from provisional to canonical. The programme convention for promoting a provisional empirical regularity to canonical status — three independent confirmations under pre-registration — is now met. The Regime 4 condition framework ($|\text{bivariate } \rho| < 0.35$ AND $\text{partial } \rho < 0$) is an empirically validated classification rule rather than a provisional zone defined by two datapoints.

Axis evolution. The headline classification chart evolves from v0.13's (bivariate $\rho \times$ decrement) view to v0.14's (bivariate $\rho \times$ partial ρ) view. The decrement axis worked when bivariate ρ was positive (so the covariate decrement was well-defined and positive); premium tea's already-negative bivariate yields a near-zero decrement that does not separate cleanly. The

partial- ρ axis classifies all three Regime 4 cases by the same condition logic that drives the H_Regime4_replication test itself.

Resolving a pre-reg wording ambiguity. Condition 2 was specified in the v0.14 pre-registration text as $\rho(\text{AI Presence, brand age})$ where the v0.13 Regime 4 framework used $\rho(\text{AI Presence, Trends})$. The canonical scoring script computes both. Both satisfy the $|\rho| < 0.35$ threshold at both waves: $\rho(\text{AI, age}) = 0.018 / -0.019$; $\rho(\text{AI, Trends}) = -0.066 / -0.134$. The empirical conclusion is robust to the ambiguity, and the v0.13 canonical interpretation (AI × Trends) is adopted going forward.

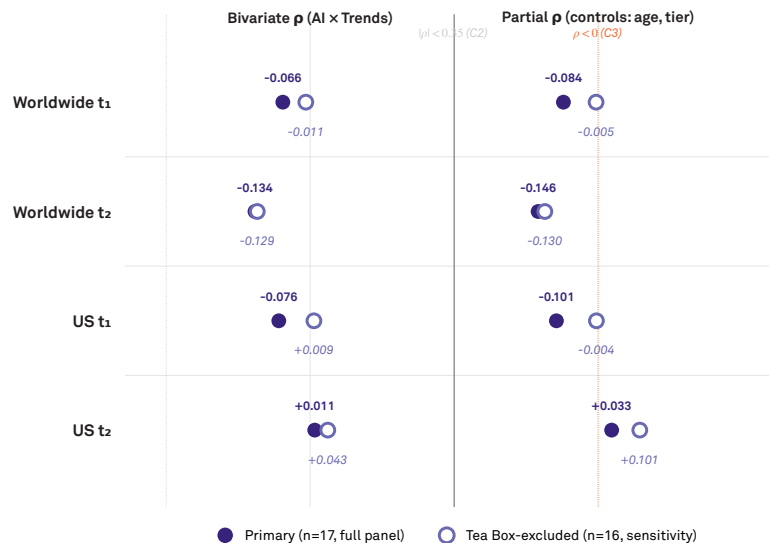
What v0.14 leaves to formalise: a sub-classification within Regime 4. The skincare-and-finance pattern (positive-to-negative migration under covariate control) differs structurally from premium tea's already-negative bivariate path, even though both satisfy the same canonical conditions. The forthcoming AIAS methodology paper will specify **Regime 4a** (migration sub-type) and **Regime 4b** (pure sub-type) thresholds analogous to those for Regimes 1–3.

FINDING 02

Tea Box-excluded sensitivity — the Regime 4 verdict is robust to specification

Primary vs Tea Box-excluded sensitivity — H_Regime4_replication robust

Each row shows bivariate ρ (left, vs C2 threshold $|\rho| < 0.35$) and partial ρ (right, vs C3 threshold $\rho < 0$) at one wave / region. Primary (indigo filled) uses the full eligible panel ($n=17$ ww, 16-17 US). Tea Box-excluded (petro open) drops the brand whose rescaled Trends signal was confounded by generic 'tea box' gift-set language. Both panels satisfy C2 + C3 at every wave; the Regime 4 verdict does not depend on Tea Box.



Source: Third System AI Presence Index v0.14 · Google Trends acquisition 2026-05-12T15:54:30Z · Pre-reg locked at v0.14-prereg (commit b0ef30a).

Figure 2 · Primary vs Tea Box-excluded sensitivity. Each row shows bivariate ρ (left, vs C2 threshold $|\rho| < 0.35$) and partial ρ (right, vs C3 threshold $\rho < 0$) at one wave / region. Primary ($n=17$, indigo filled) uses the full eligible panel; Tea Box-excluded ($n=16$, petro open) drops the brand whose rescaled Trends signal was confounded by generic 'tea box' gift-set language. Both panels satisfy C2 + C3 at every wave; the Regime 4 verdict does not depend on Tea Box.

The Regime 4 verdict does not depend on Tea Box. One of the 17 eligible primary brands in the premium-tea panel is the brand 'Tea Box' (mainstream-premium tier), whose Trends rescaled mean (166 worldwide at t_2 , 406 US at t_2) is anomalously high relative to its AI Presence rate (2.1%). The cause: the generic phrase 'tea box' captures gift-set search volume unrelated to the brand entity. The Tea Box-excluded sensitivity drops this brand and re-runs the three pre-registered conditions on $n = 16$.

All three conditions still hold at both waves. Bivariate Spearman $\rho(\text{AI} \times \text{Trends})$ is -0.011 at t_1 and -0.129 at t_2 — both well within the C2 threshold of $|\rho| < 0.35$. Partial ρ after age + premium-tier control is -0.005 at t_1 and -0.130 at t_2 — both satisfying C3 partial $\rho < 0$. The sample size at $n = 16$

remains above the C1 floor of 12.

The primary and sensitivity panels arrive at structurally identical conclusions. The chart at right shows bivariate ρ and partial ρ side-by-side across all four wave/region combinations, primary ($n = 17$) versus Tea Box-excluded ($n = 16$). At every wave and every region, both panels satisfy both the C2 and C3 thresholds. The verdict is robust to the inclusion or exclusion of any single brand whose Trends measurement may have been confounded by generic-phrase capture.

What this means methodologically. The pivot-rescaling Trends acquisition occasionally captures generic-phrase search volume in addition to brand-specific search; the Tea Box case illustrates the failure mode in a category where it has

greatest leverage (a brand whose name is also a category-level descriptive phrase). For the v0.14 Regime 4 verdict the leverage is small enough not to flip the result, but the operational lesson generalises: pre-registered sensitivity tests against generic-phrase contamination are a useful component of the AIAS Protocol going forward.

US-region sensitivity also confirms. The US-region subset re-runs the H_Regime4_replication conditions: bivariate $\rho = -0.076 / +0.011$; partial $\rho = -0.101 / +0.033$. The US-region

partial at t_2 sits very close to zero ($+0.033$), on the boundary of C3, but the worldwide-region verdict — the primary analysis — carries definitively.

Premium tea joins skincare and finance as Regime 4 datapoints with sensitivity tests that confirm the verdict rather than narrow it. The condition framework has accumulated robustness across both within-version (multiple sensitivity panels) and cross-version (three independent categories) replication.

FINDING 03

Per-category construct validity — six categories on common axes

The six-category panel shows three structurally different relationships between AI Presence and consumer-search rank. v0.14 measures premium tea fresh alongside the five-category panel from v0.13 (project management software, premium running shoes, premium olive oil, premium facial skincare, personal-finance apps). The chart below places each category at its bivariate worldwide ρ (indigo), partial worldwide ρ (petro), and bivariate US ρ (grey, sensitivity) for both waves. The premium-tea row is the v0.14 addition.

Regime 1 — PM software. Bivariate $\rho = 0.506 / 0.482$, partial $\rho = 0.417 / 0.434$. The marginal-direct pattern reproduces in v0.14 within sampling error. v0.11 / v0.12 / v0.13 / v0.14 confirm PM software at the lower edge of construct-validity confirmation, with covariate decrement small enough (~ 0.07) to leave a positive residual.

Regime 2 — running shoes. Bivariate $\rho = 0.808 / 0.786$, partial $\rho = 0.466 / 0.488$. H1 confirms decisively at both waves. The substantial covariate decrement (~ 0.34) reflects age-driven joint visibility — older heritage brands rank highly in both AI Presence and Trends because their age has produced both training-corpus exposure and search familiarity. The residual partial after age + tier control sits at PM-software's marginal-direct level.

Regime 3 — olive oil. $n = 8$ worldwide and $n = 7$ US, both below the hard floor of 10. Routes to descriptive-only per

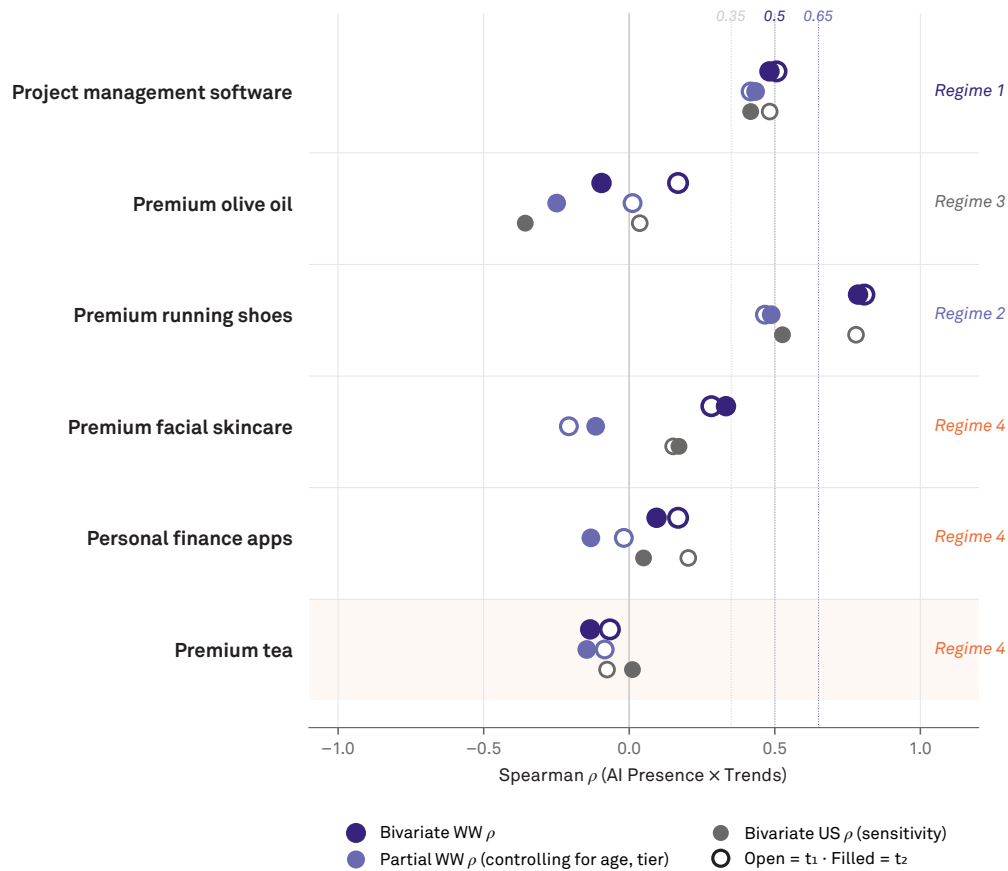
pre-reg §3.6a. The v0.12 Category-Scale Mismatch finding (7 of 15 matched-subset brands with AI Presence $\uparrow$ 5% and Trends below display threshold) carries forward unchanged. The category is not refuted; it sits outside the H1–H4 inference path.

Regime 4 — skincare, finance, premium tea. Three datapoints, three positions within the regime. Skincare sits at the boundary of Regime 1 (bivariate $0.28 / 0.33$, partial $-0.21 / -0.12$) where the covariates absorb the entire bivariate signal and reveal a slight negative residual. Finance sits unambiguously inside Regime 4 (bivariate $0.17 / 0.09$, partial $-0.16 / -0.29$). Premium tea sits in the pure-form region (bivariate $-0.07 / -0.13$, partial $-0.08 / -0.15$) — the cleanest case to date.

The structural heterogeneity within Regime 4 is the next methodological question. A formal Regime 4 sub-classification distinguishes a migration sub-region (Regime 4a: positive bivariate that flips negative under covariate control — skincare, finance) from a pure sub-region (Regime 4b: already-negative bivariate; partial similarly negative — premium tea). The unified condition signature ($|\text{bivariate } \rho| < 0.35 \text{ AND } \text{partial } \rho < 0$) is satisfied by both sub-types and serves as the canonical operational test; the sub-classification awaits additional datapoints in each sub-region.

Per-category construct validity — v0.13 plus v0.14 (premium tea)

Bivariate WW ρ (indigo) measures the AI Presence $\times$ Trends rank co-movement. Partial WW ρ (petro) controls for brand age and tier. Bivariate US ρ (grey) is the US-region sensitivity. Open markers = t_1 ; filled = t_2 . Premium tea (v0.14, highlighted row) joins skincare and finance in the Regime 4 cluster.



Source: Third System AI Presence Index v0.14 · Google Trends acquisition 2026-05-12T15:54:30Z · Pre-reg locked at v0.14-prereg (commit b0ef30a).

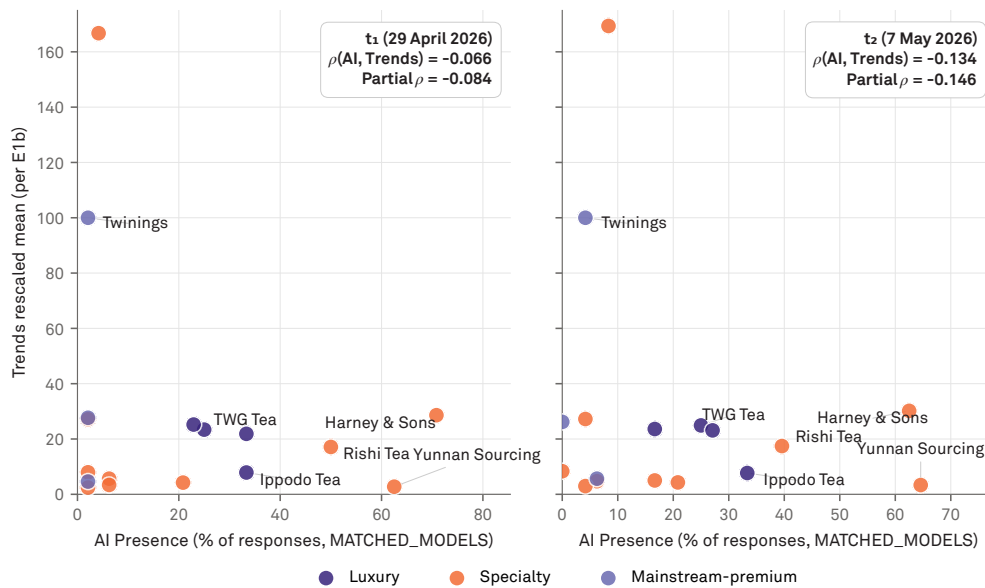
Figure 3 · Per-category construct validity, v0.13 + v0.14. Six categories with bivariate WW ρ (indigo), partial WW ρ (petro), and bivariate US ρ (grey) at t_1 (open) and t_2 (filled). Running anchors the upper end (≈ 0.80); PM software at the marginal-direct boundary (≈ 0.50); skincare and finance show the canonical Regime 4 pattern (positive bivariate, negative partial); premium tea (highlighted row) is the v0.14 addition with bivariate already negative — the cleanest Regime 4 case yet.

FINDING 04

Within-category structure — AI surfaces specialty and Asian-tradition brands

Premium tea — AI Presence × Trends, per-brand scatter

Each point is a brand at one wave; eligible (E1a + E1b) brands only. Coloured by premium_tier. Top AI Presence brands (Harney & Sons, Yunnan Sourcing, Ippodo) anchor the upper-x; the Trends pivot Twinings is lower-x, illustrating the Regime 4 decoupling — AI Presence ranks specialty/Asian brands that the Trends signal does not surface.



Source: Third System AI Presence Index v0.14 · Google Trends acquisition 2026-05-12T15:54:30Z · Pre-reg locked at v0.14-prereg (commit b0ef30a).

Figure 4 · Premium tea, per-brand AI Presence × Trends rescaled at t_1 and t_2 . Each point is an eligible (E1a + E1b) brand; coloured by premium_tier (luxury / specialty / mainstream-premium). Harney & Sons, Yunnan Sourcing, and Ippodo Tea anchor the high-AI cluster; Twinings — the Trends pivot — sits in the low-AI cluster, making the Regime 4 decoupling visible: AI Presence ranks specialty and Asian brands that the Trends signal does not surface.

The within-category structure of premium tea makes the Regime 4 substrate visible. Top AI Presence brands at v0.14, averaged across both waves on the matched-model subset, are Harney & Sons (59%), Yunnan Sourcing (57%), Ippodo Tea (49%), Rishi Tea (41%), and TWG Tea (37%). This is a mix of US specialty (Harney, Rishi), Asian-tradition specialty (Yunnan Sourcing, Ippodo), and Singaporean luxury (TWG).

The Trends pivot, Twinings, sits at AI Presence 6.2% — the starkest divergence between AI and consumer-search rankings observed in the v0.14 panel. The chart at right shows each eligible brand's position in (AI Presence × Trends rescaled mean) space at both waves, coloured by premium tier. Twinings anchors the high-Trends / low-AI corner; the top AI brands anchor the high-AI / low-Trends corner. The

decoupling is geometrically visible.

What this reveals about the substrate. Matched-model LLMs (Claude Sonnet 4.6 + GPT-5.4-mini) surface specialty tea expertise — loose-leaf curation, sourcing knowledge, Asian-tradition matcha and pu-erh varieties — that English-language consumer search does not concentrate on. This is not a measurement artefact: the LLM training corpora include substantial coverage of specialty tea writing (Ippodo's matcha guides, Yunnan Sourcing's pu-erh sourcing notes, Harney's blending tradition) that is not reflected in Trends' aggregate search-volume signal.

Twinings's position is the diagnostic. Twinings dominates English-language tea search by an order of magnitude over

any single specialty brand, but does not dominate LLM responses to category-recommendation prompts. The asymmetry has a clean reading: consumer search is structured around the brands consumers already know (Twinings, mass-market familiarity); LLM responses are structured around the brands the model has seen discussed substantively in training corpora (specialty curation, Asian-tradition expertise). Premium tea's Regime 4 signature is the geometric consequence of this asymmetry.

The category-specific substrate may not generalise.

Skincare and finance share the Regime 4 condition outcome but reach it through a different mechanism — large incumbent brands with high age + tier values that LLMs surface because of training-data persistence, where covariate control reveals the underlying decoupling. Premium tea's

mechanism is the specialty / incumbent asymmetry. The canonical Regime 4 should not be interpreted as a single underlying mechanism — it is an empirical co-occurrence of two conditions that admits multiple substrates.

v0.15 registry candidates surfaced in v0.14. Four high-mention brands not in the v0.14 registry were observed: Mariage Frères (144 mentions across the wave panel), Palais des Thés (82), White2Tea (83), and Rare Tea Company (74). Inclusion in v0.15 will expand the panel from $n = 17$ to approximately $n = 25-27$, with Mariage Frères likely entering the top-5 AI Presence ranks (its mention count exceeds Twinings's). The Regime 4 verdict is unlikely to invert under expansion — the additional brands amplify the specialty / luxury cluster that already drives the signature.

Limitations

Single category at v0.14. The replication test is in one category (premium tea). The strength of the result rests on cross-version triangulation with v0.13's skincare and finance findings rather than on within-version replication. Future phases (v0.15 registry expansion; v0.16+ designed-for-test in additional categories) will test whether the Regime 4 signature persists across more datapoints and whether the Regime 4a / 4b sub-classification stabilises with additional datapoints in each sub-region.

Two-wave short window. Both waves are within 8 days of each other ($t_1 = 29$ April 2026; $t_2 = 7$ May 2026). The cross-wave stability finding (H2 confirmed) is short-window. Longer-horizon stability of the Regime 4 signature — month-to-month or quarter-to-quarter — is not tested at v0.14.

Matched-model subset (two models). AI Presence is computed across Claude Sonnet 4.6 + GPT-5.4-mini at status = ok. The Tri-System framework's three-mode response taxonomy (Brand mode / Component mode / Authority mode) suggests inter-model variation in AI Presence may be substantial; the matched-subset constraint factors out this variation but does not address it. Future programme phases will report results across additional model families.

Single external validator (Google Trends). The construct-validity test uses Google Trends as the sole external consumer-search reference. A multi-validator design — testing AI Presence simultaneously against search interest, social-media mention rates, retail sales data where available, and consumer-survey aided-recall measures — would generalise the construct-validity claim from a single-validator finding to a multi-validator finding. This is the medium-term goal of the AIAS programme (Phase 3).

Registry coverage gap. Four high-mention brands not in the v0.14 registry were observed in LLM responses: Mariage Frères (144 wave-2 mentions), Palais des Thés (82), White2Tea (83), Rare Tea Company (74). At the matched-model subset's totals these would likely enter the top-5 AI Presence ranks (Mariage Frères ahead of Twinings, possibly ahead of TWG Tea). Their absence from v0.14 reflects registry-construction timing — the registry was locked before mention frequencies were observed. The Regime 4 verdict is unlikely to invert under their inclusion (they amplify the specialty / luxury cluster that already drives the signature), but per-brand rankings should be read against this caveat. v0.15 expansion is queued.

Tea Box query ambiguity. The pivot-rescaling Trends acquisition for the Tea Box brand captured generic-phrase search volume ("tea box" as gift-set descriptor) in addition to brand-specific search. The Tea Box-excluded sensitivity panel (§3.2 in the SSRN paper; H_Regime4_replication second row in the scoring table above) addresses this directly: all three pre-registered conditions hold at $n = 16$ with Tea Box dropped. The primary verdict is robust to Tea Box's inclusion or exclusion; the failure mode is documented and addressed in the AIAS Protocol v1.2 specification (Phase B disambiguation step).

What's Next

v0.15 — premium tea registry expansion. Add Mariage Frères (144 wave-2 mentions), Palais des Thés (82), White2Tea (83), Rare Tea Company (74), plus approximately 6 additional high-mention specialty brands surfaced in v0.14's LLM responses (What-Cha, Upton Tea Imports, Kettl, Yunomi, Den's Tea, Seven Cups, Camellia Sinensis). Re-run H_Regime4_replication on the expanded $n \approx 25\text{--}27$ panel. Pre-registered prediction: the Regime 4 verdict persists; expansion amplifies the specialty / Asian-tradition cluster that drives the signature.

AIAS methodology paper. Formalise the four-regime taxonomy with threshold-precise definitions analogous to those for Regimes 1–3. Specify the Regime 4a (migration sub-type — skincare, finance) / Regime 4b (pure sub-type — premium tea) sub-classification once a fourth datapoint is in hand. Working title: *Measuring AI Availability: Methodological Notes from the AIAS Protocol*. Queue position now open since the Tri-System Brand Growth paper is at MSI Working Paper Series.

AIAS Protocol v1.2. Increment from v1.1 to incorporate regime classification as a category-level routing step in the canonical measurement pipeline. Categories classifying to Regime 4 receive category-specific commentary on the operational substrate, since the unified condition signature admits multiple substrates — LLM training-corpus exposure asymmetry for premium tea; age-and-tier confounds for skincare and finance. Phase B generic-phrase disambiguation step formalises the Tea Box failure-mode lesson.

Phase 3 (AIAS components 2–6). Construct validity established for AI Presence in v0.11 through v0.14 is a precondition for measurement work on the five remaining AIAS components: Ranking, Consistency, Coverage, Grounding, Sentiment. The four-regime taxonomy suggests each component will require its own per-category construct-validity profile rather than a uniform cross-category correlation with any single external proxy.

External brand-tracking validation. Phase 3 will test AI Presence against external brand-tracking data (Kantar BrandZ, YouGov BrandIndex, brand health tracker panels). The four-regime taxonomy provides pre-registered predictions: brands in Regime 1 should show stronger AI Presence $\times$ brand-tracking correlation than brands in Regime 4. The cross-system validity test will sharpen the construct-validity claim from a Trends-only finding to a multi-validator finding.

Hypothesis Scoring

All thresholds and tests locked at v0.14-prereg (commit b0ef30a, 12 May 2026 UTC) prior to any LLM acquisition or Google Trends acquisition call against the wave windows. Per-category pivot exemption from E1b applied per pre-reg §5.1 (sd = 0 by pivot construction). The primary analysis is the worldwide-region cell at both waves; the Tea Box-excluded sensitivity and the US-region subset serve as pre-registered robustness panels.

H	Pre-registered prediction	Result	Status
H_Regime4_replication (primary)	$n \nless 12$ AND $ bivariate\ p(AI, Trends) < 0.35$ AND $partial\ p(AI, Trends \mid age, tier) < 0$; both waves WW	$n = 17$; bivariate $p = -0.066 / -0.134$; partial $p = -0.084 / -0.146$ — all three conditions satisfied	CONFIRMED
H_Regime4_replication (Tea Box-excluded)	Same three conditions on $n = 16$ (Tea Box dropped); sensitivity	$n = 16$; bivariate $p = -0.011 / -0.129$; partial $p = -0.005 / -0.130$ — verdict robust	CONFIRMED
H1 (per-category)	Spearman $\rho > 0.5$ AND $p_{t1} < 0.05$, both waves	$\rho_{t_1} = -0.066$; $\rho_{t_2} = -0.134$ — well below 0.5 by sign and magnitude	FALSIFIED
H2 (cross-wave stability)	$ \Delta p \nless 0.15$ between t_1 and t_2	$ \Delta p = 0.068$ — well within stability threshold	CONFIRMED
H3 (leader ship-zone subset)	AI top-3 Trends top-5, both waves	0 / 3 brands overlap at both waves (Harney / Yunnan / Ippodo absent from Trends top-5)	FALSIFIED
H4 (partial -Spearman)	Partial $\rho > 0.5$ (age + premium_tier), both waves	partial $p = -0.084 / -0.146$ (covariate-saturated, negative-residual)	FALSIFIED
H7(regime classification)	Matches Regime 1 / 2 / 3 (the pre-registered set)	Matches none — Regime 4 candidate; confirmed under H_Regime4_replication	FALSIFIED (productive)

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. H_Regime4_replication is the v0.14 primary hypothesis; H1–H4 are per-category construct-validity tests carried forward from the programme convention (their falsification pattern is the structural complement of H_Regime4_replication confirmation). H7 tests whether premium tea classifies cleanly into one of the three pre-registered regimes; its productive falsification is the route by which premium tea enters Regime 4.

Primary hypothesis (v0.14). Three pre-registered conditions tested at both worldwide waves: (C1) $n \nless 12$ eligible brands; (C2) $|bivariate\ p(AI\ Presence, Trends)| < 0.35$; (C3) $partial\ p(AI\ Presence, Trends \mid age, premium_tier) < 0$. All three satisfied at t_1 and t_2 : $n = 17$ (above C1 floor of 12); bivariate $p = -0.066 / -0.134$ (within C2 threshold by an order of magnitude); partial $p = -0.084 / -0.146$ (satisfying C3). **CONFIRMED.** Three

independent confirmations across three categories (premium facial skincare and personal-finance apps in v0.13; premium tea in v0.14) elevate Regime 4 from provisional to canonical.

Tea Box-excluded sensitivity confirms. One eligible brand ("Tea Box", mainstream-premium tier) carries a Trends rescaled mean (166 worldwide t_2 , 406 US t_2) anomalously high relative to its AI Presence (2.1%) because the generic phrase

"tea box" captures gift-set search volume. Sensitivity panel drops it ($n = 16$): bivariate $\rho = -0.011 / -0.129$; partial $\rho = -0.005 / -0.130$. All three pre-registered conditions still satisfied. The Regime 4 verdict is robust to the inclusion or exclusion of any single brand whose Trends measurement may have been confounded by generic-pharse capture.

H1 (bivariate Spearman $\rho > 0.5$ both waves) FALSIFIED. $\rho_{t_1} = -0.066$; $\rho_{t_2} = -0.134$. Falsified by both sign and magnitude¹—bivariate ρ is already negative at both waves, the cleanest H1 falsification in the programme to date and the operational signature of Regime 4b (pure sub-type) as distinct from skincare and finance's Regime 4a (migration sub-type).

H2 (cross-wave stability $|\Delta\rho| \leq 0.15$) CONFIRMED. $|\Delta\rho| = 0.068$ between t_1 and t_2 (eight-day interval). The Regime 4 signature is stable across the inter-wave window. Whatever the matched-model LLMs' training-corpus exposure captures, it captures the same thing at both measurement points; the asymmetry between AI Presence and consumer-search rank is not a wave-specific artefact.

H3 (top-3 AI $\square$ top-5 Trends both waves) FALSIFIED. 0 of 3 brands overlap at both waves. AI top-3 across both waves: Harney & Sons, Yunnan Sourcing, Ippodo Tea. Trends top-5 worldwide: Twinings (pivot), Republic of Tea, Whittard of

Chelsea, Tea Pigs, Lupicia. The leadership zones are entirely disjoint — the cleanest H3 falsification in the programme to date and the geometric expression of the within-category decoupling that drives the Regime 4 signature.

H4 (partial Spearman $\rho > 0.5$ both waves) FALSIFIED. Partial $\rho = -0.084 / -0.146$ after controlling for brand_age_years and premium_tier ordinal. Falsified by sign and magnitude. The partial-correlation machinery confirms the bivariate decoupling is not concealed by age or tier confounds: the covariate decrement (bivariate – partial ≈ 0.02) is near-zero, distinct from v0.13 skincare's decrement of ~ 0.49 and finance's ~ 0.18 . Premium tea reaches Regime 4 partial signature via an already-negative bivariate, not via covariate dissolution.

H7 (clean classification to Regime 1 / 2 / 3) FALSIFIED productively. Premium tea matches none of the three pre-registered regimes. Bivariate $\rho = -0.066 / -0.134$ — below Regime 1's lower bound of 0.35 and above Regime 2's lower bound of 0.65 in absolute terms would still place it nowhere in the pre-registered set, and $n = 17 \blacktriangle 10$ puts it outside Regime 3 (Scale-mismatch via §3.6a). The productive falsification is the route by which premium tea enters Regime 4, joining skincare and finance in the canonical lower-left quadrant of (bivariate $\rho \times$ partial ρ) space.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

Authored by

Pablo Ulpiano Gonzalez Castro

Principal Researcher, Third System

Faculty, MPS Branding Program, School of Visual Arts

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *Four Empirical Regimes in AI-Mediated Brand Visibility: A Five-Category Replication of the v0.13 Regime 4 (Covariate-saturated Weak) Finding in a Designed-for-Test Category — Premium Tea — AI Presence Index v0.14*. Third System.
thirdsystem.ai/v14-premium-tea-regime4-replication (SSRN forthcoming)

Underlying datasets

OSF project [ec6wh](https://osf.io/ec6wh/), /v14/. Inputs: fresh LLM acquisition (288 calls across two waves on Claude Sonnet 4.6 + GPT-5.4-mini matched-model subset, status = ok); Google Trends raw responses (8 bundles × 2 waves × 2 regions, locked at acquisition timestamp 2026-05-12T15:54:30Z); Phase A pivot validation (Twinings, mean 88.14, CV 9.13%); Phase B topic-ID resolution and bare-canonical-query outputs; Phase B-alternates activation logs (Yunnan Sourcing); premium tea registry (brands_premium_tea.json, v3 schema, 25 brands organised by premium_tier: 6 luxury, 3 specialty, 16 mainstream-premium); brand-age verified table; scoring outputs (canonical_scoring.json, h_regime4_replication.csv, h7_regime_classification.csv, per_brand_paired.csv); 4 chart PDFs (regime4 canonical, primary vs sensitivity, per-category construct-validity comparison, premium tea per-brand scatter); build scripts; this report; and the matching SSRN working paper.

Companion SSRN working paper (Gonzalez Castro 2026, SSRN 6755621). Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163); v0.11 PM Software × Trends Construct Validity Pilot (SSRN 6745040); v0.12 Three Empirical Regimes (SSRN 6748341); v0.13 Four Empirical Regimes — Five-Category Construct-Validity Expansion (SSRN 6750498).

Methodology log: v0.14 follows AIAS Presence Measurement Protocol v1.1 (unchanged from v0.9 through v0.13). Pre-registration locked at git tag v0.14-prereg (commit b0ef30a) on 12 May 2026 UTC prior to LLM acquisition. Trends acquisition session UTC timestamp 2026-05-12T15:54:30Z. DEVIATIONS.md Entry 1 documents the Phase B bare-canonical-query methodology amendment (pytrends topic-ID errors — Republic of Tea matching to an Irish football team, Makaibari to a hotel — necessitated abandoning topic-ID resolution); Entry 2 documents Yunnan Sourcing alternate activation after Wang De Chuan and In Pursuit of Tea failed both solo and bundled-E5 padded-resolution checks. No pre-registered hypothesis, threshold, or routing rule was modified. The canonical scoring script (score_v14.py) is locked at git commit e71e135..

Inquiries: hello@thirdsystem.ai